

ENVIRONMENT SETUP

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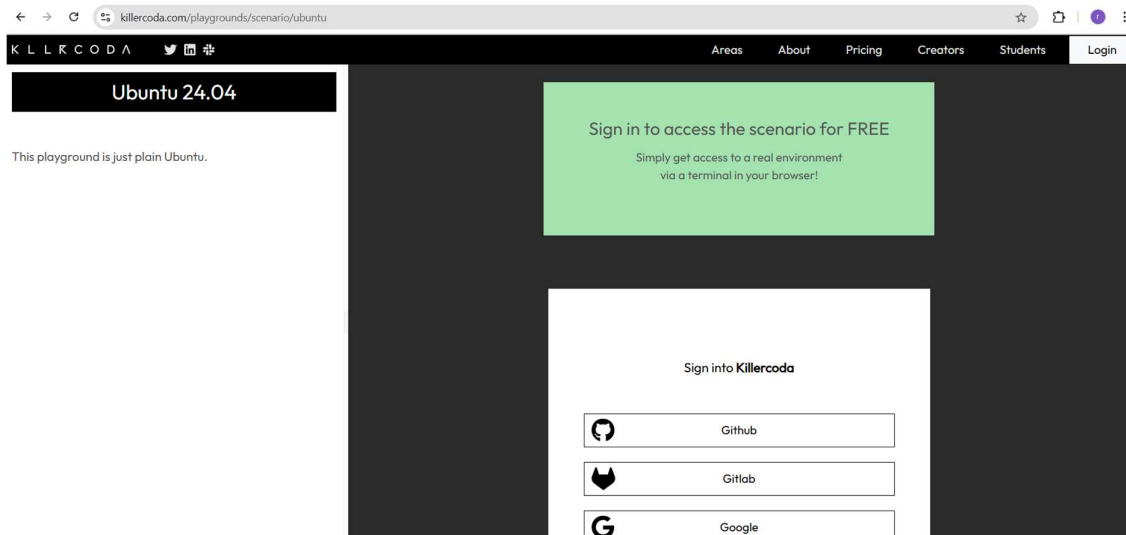
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1.Podman Setup Guide.

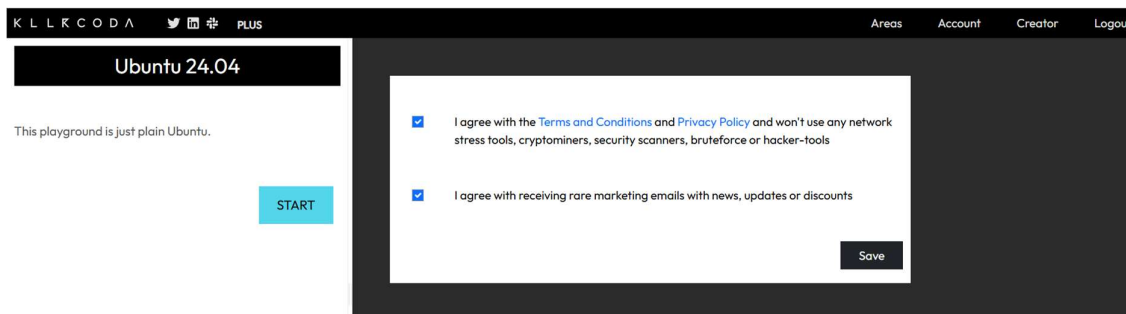
For our practical , we'll be using a fantastic online platform called **Killercoda**. Think of Killercoda as your own personal, ready-to-use virtual computer, pre-loaded with all the tools you need for our DevOps course, including Podman!

This means you don't need to worry about installing Podman on your own laptop, or dealing with any setup issues. Just open your web browser, and you're ready to go!

1. For the Podman practices, we will use ‘killer coda interactive environments”. For this, go to the link <https://killercoda.com/playgrounds> and click on “ubuntu 24.04”.
2. Click on “I agree” . then , following screen appears



3. Sign in into using your Google Account



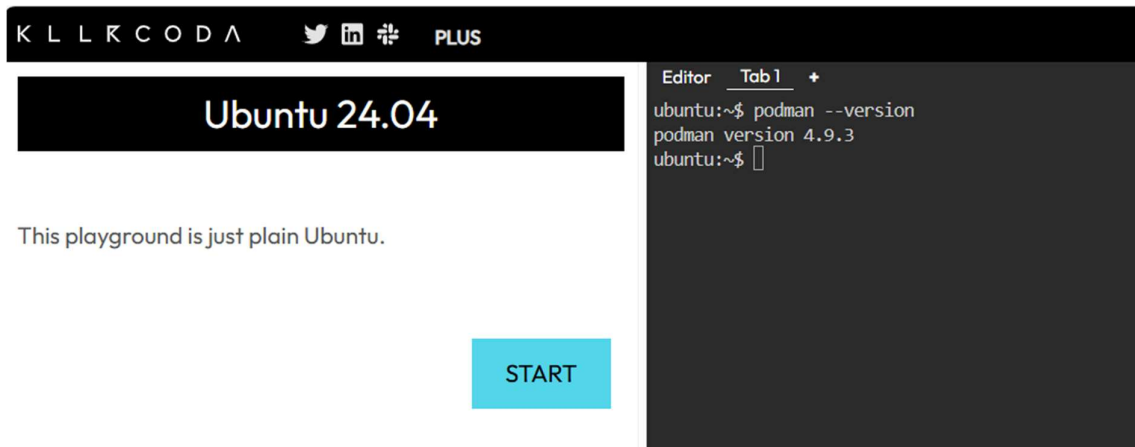
4. Check on “I Agree” checkbox and then click on save.



5. Here , you will notice, that it starts plain ubuntu and at right side, you will see the timer for 60 minutes.

Note : The session will expire after 60 minutes

6. You will notice that the Podman is already installed in it. So there Is no need to install the Podman on ubuntu. You need to just type the command and check its functioning



7. Go to areas tab and search and select for podman

Explore

Profiles

Scenarios

Podman

Manage pods, containers, and container images.

Podman is a daemonless container engine for developing, managing, and running OCI Containers on your Linux System. Containers can either be run as root or in rootless mode. [MAINTAINERS] You can manage and change the scenarios here by creating pull requests on <https://github.com/killercoda/scenarios-podman>

1 Scenarios

2. Installing WSL Ubuntu and Setting Up Jenkins

Overview

Jenkins is an open-source software package for continuous software development. It is used to automate parts of the build, testing, delivery, and deployment of applications.

Jenkins is based on Java and helps with every part of the software development process.

Prerequisites for WSL Installation:

1. Windows 10 version 2004 or higher, or Windows 11.
2. Virtual Machine Platform and WSL features must be enabled.
3. Admin access to run PowerShell commands.
4. Internet connection to download Linux distro.
5. Virtualization must be enabled in BIOS/UEF

Tasks:

1. Open PowerShell as Administrator.
2. Run the command to enable WSL:

```
wsl --install
```

This installs the default Ubuntu distro.

3. Restart your system if prompted.
4. Search for "Ubuntu" in the Start menu and launch it.
5. Set up a username and password when prompted.
6. Update and upgrade Ubuntu packages:

```
sudo apt update && sudo apt upgrade -y
```

7. Install Java and required utilities:

```
sudo apt install -y gnupg curl software-properties-common openjdk-17-jdk
```

8. Verify Java installation:

```
java -version
```

9. Add Jenkins repository and key:

```
curl -fsSL https://pkg.jenkins.io/debian/jenkins.io-2023.key | sudo tee \
/usr/share/keyrings/jenkins-keyring.asc > /dev/null
```

```
echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian binary/ | \
sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
```

```
richa_ubuntu@DESKTOP-QUFRMJF:~$ sudo apt update
Hit:1 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:2 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:3 https://ppa.launchpadcontent.net/ansible/ansible/ubuntu noble InRelease
Hit:4 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:5 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
35 packages can be upgraded. Run 'apt list --upgradable' to see them.
richa_ubuntu@DESKTOP-QUFRMJF:~$ curl -fsSL https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key | sudo tee /usr/share/
keyrings/jenkins-keyring.asc > /dev/null
richa_ubuntu@DESKTOP-QUFRMJF:~$ echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] https://pkg.jenkins.io/debian
-stable binary/ | \
sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
```

10. Update package list and install Jenkins:

sudo apt update

```
richa_ubuntu@DESKTOP-QUFRMJF:~$ sudo apt update
Ign:1 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:2 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:3 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Get:4 https://pkg.jenkins.io/debian-stable binary/ Packages [28.8 kB]
Hit:5 http://archive.ubuntu.com/ubuntu noble InRelease
Hit:6 http://archive.ubuntu.com/ubuntu noble-updates InRelease
Hit:7 https://ppa.launchpadcontent.net/ansible/ansible/ubuntu noble InRelease
Hit:8 http://security.ubuntu.com/ubuntu noble-security InRelease
Hit:9 http://archive.ubuntu.com/ubuntu noble-backports InRelease
Fetched 31.6 kB in 2s (18.6 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
35 packages can be upgraded. Run 'apt list --upgradable' to see them.
```

sudo apt install jenkins -y

```
richa_ubuntu@DESKTOP-QUFRMJF:~$ sudo apt install jenkins -y
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  net-tools
The following NEW packages will be installed:
  jenkins net-tools
0 upgraded, 2 newly installed, 0 to remove and 35 not upgraded.
Need to get 95.0 MB of archives.
After this operation, 97.6 MB of additional disk space will be used.
Get:2 http://archive.ubuntu.com/ubuntu noble/main amd64 net-tools amd64 2.10-0.1ubuntu4 [204 kB]
Get:1 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.492.3 [94.8 MB]
47% [1 jenkins 44.2 MB/94.8 MB 47%]
Fetched 95.0 MB in 1min 54s (832 kB/s)
Selecting previously unselected package net-tools.
(Reading database ... 68363 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-0.1ubuntu4_amd64.deb ...
Unpacking net-tools (2.10-0.1ubuntu4) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins_2.492.3_all.deb ...
Unpacking jenkins (2.492.3) ...
Setting up net-tools (2.10-0.1ubuntu4) ...
Setting up jenkins (2.492.3) ...
Created symlink /etc/systemd/system/multi-user.target.wants/jenkins.service → /usr/lib/systemd/system/jenkins.service.
Processing triggers for man-db (2.12.0-4build2) ...
```

11. Verify Jenkins installation

dpkg -l | grep Jenkins

```
richa_ubuntu@DESKTOP-QUFRMJF:~$ dpkg -l | grep jenkins
ii  jenkins      2.492.3      all          Jenkins is the leading open source
automation server supported by a large and growing community of developers, testers, designers and other people interested
in continuous integration, continuous delivery and modern software delivery practices. Built on the Java Virtual Machine
(JVM), it provides more than 1,800 plugins that extend Jenkins to automate with practically any technology software delive
ry teams use. In 2022, Jenkins reached 300,000 known installations making it the most widely deployed automation server.
```

12. Start , enable the jenkins service and the check the service status of jenkins

`sudo service jenkins start`

`sudo systemctl enable jenkins`

`sudo systemctl status jenkins`

```
richa_ubuntu@DESKTOP-QUFRMJF:~$ sudo service jenkins start
richa_ubuntu@DESKTOP-QUFRMJF:~$ sudo systemctl enable jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
richa_ubuntu@DESKTOP-QUFRMJF:~$ sudo service jenkins status
• jenkins.service - Jenkins Continuous Integration Server
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Fri 2025-04-11 07:36:34 UTC; 3min 42s ago
     Main PID: 4550 (java)
        Tasks: 46 (limit: 2236)
      Memory: 570.9M ()
      CGroup: /system.slice/jenkins.service
              └─4550 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --

Apr 11 07:36:15 DESKTOP-QUFRMJF jenkins[4550]: 8a0d2f200101419c8791390413f87ad2
Apr 11 07:36:15 DESKTOP-QUFRMJF jenkins[4550]: This may also be found at: /var/lib/jenkins/sec
Apr 11 07:36:15 DESKTOP-QUFRMJF jenkins[4550]: *****
Apr 11 07:36:15 DESKTOP-QUFRMJF jenkins[4550]: *****
Apr 11 07:36:15 DESKTOP-QUFRMJF jenkins[4550]: *****
Apr 11 07:36:34 DESKTOP-QUFRMJF jenkins[4550]: 2025-04-11 07:36:34.627+0000 [id=31] INFO
```

13. Check the jenkins on port in wsl

`cat /etc/default/jenkins`

Look for the line:

`HTTP_PORT=8080`

14. Open a browser and navigate to:

<http://localhost:8080>

The image shows the 'Unlock Jenkins' screen from the Jenkins web interface. At the top, it says 'Getting Started'. The main heading is 'Unlock Jenkins'. Below it, a paragraph explains that a password has been written to the log and a file on the server. The file path is shown in red text: `/Users/marko/.jenkins/secrets/initialAdminPassword`. A prompt asks the user to copy the password from either location and paste it below. There is a text input field labeled 'Administrator password'. A 'Continue' button is at the bottom right. A faint background image of the Leaning Tower of Pisa is visible on the right side.

Getting Started

Unlock Jenkins

To ensure Jenkins is securely set up by the administrator, a password has been written to the log (not sure where to find it?) and this file on the server:

`/Users/marko/.jenkins/secrets/initialAdminPassword`

Please copy the password from either location and paste it below.

Administrator password

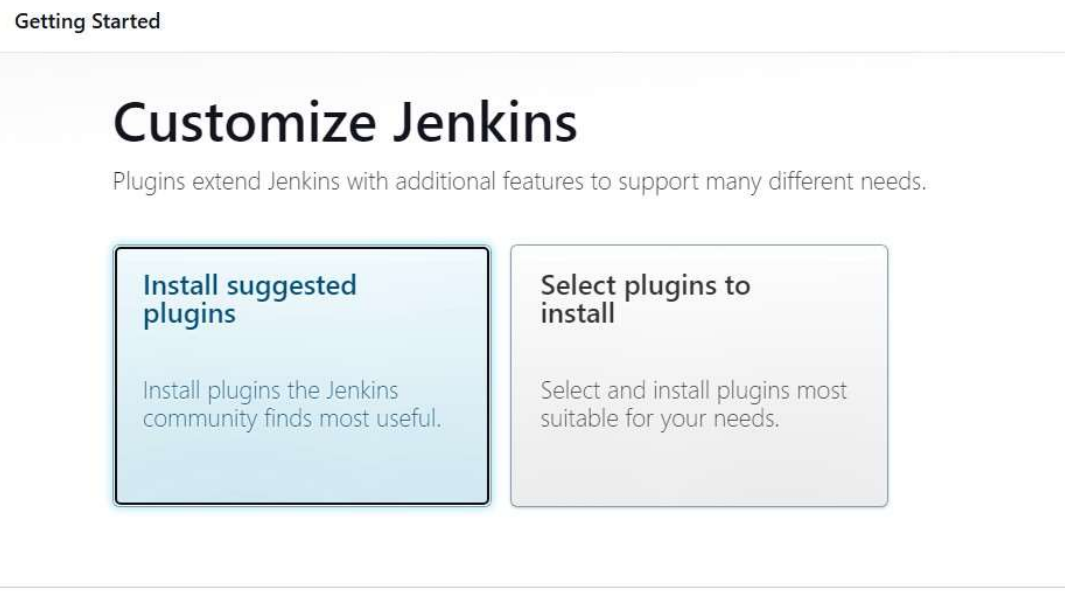
Continue

15. To unlock Jenkins for the first time in WSL , Retrieve the initial admin password:

`sudo cat /var/lib/jenkins/secrets/initialAdminPassword`

Copy the password displayed in the terminal and Paste the password into the browser prompt.

16. Follow the setup wizard to install suggested plugins and create an admin user.

The image shows the 'Customize Jenkins' screen from the Jenkins web interface. At the top, it says 'Getting Started'. The main heading is 'Customize Jenkins'. Below it, a paragraph says 'Plugins extend Jenkins with additional features to support many different needs.' There are two main options presented in boxes: 'Install suggested plugins' (highlighted with a blue border) and 'Select plugins to install'. The first option includes the text 'Install plugins the Jenkins community finds most useful.' The second option includes the text 'Select and install plugins most suitable for your needs.' Below these options are two horizontal lines for additional input or notes.

Getting Started

Customize Jenkins

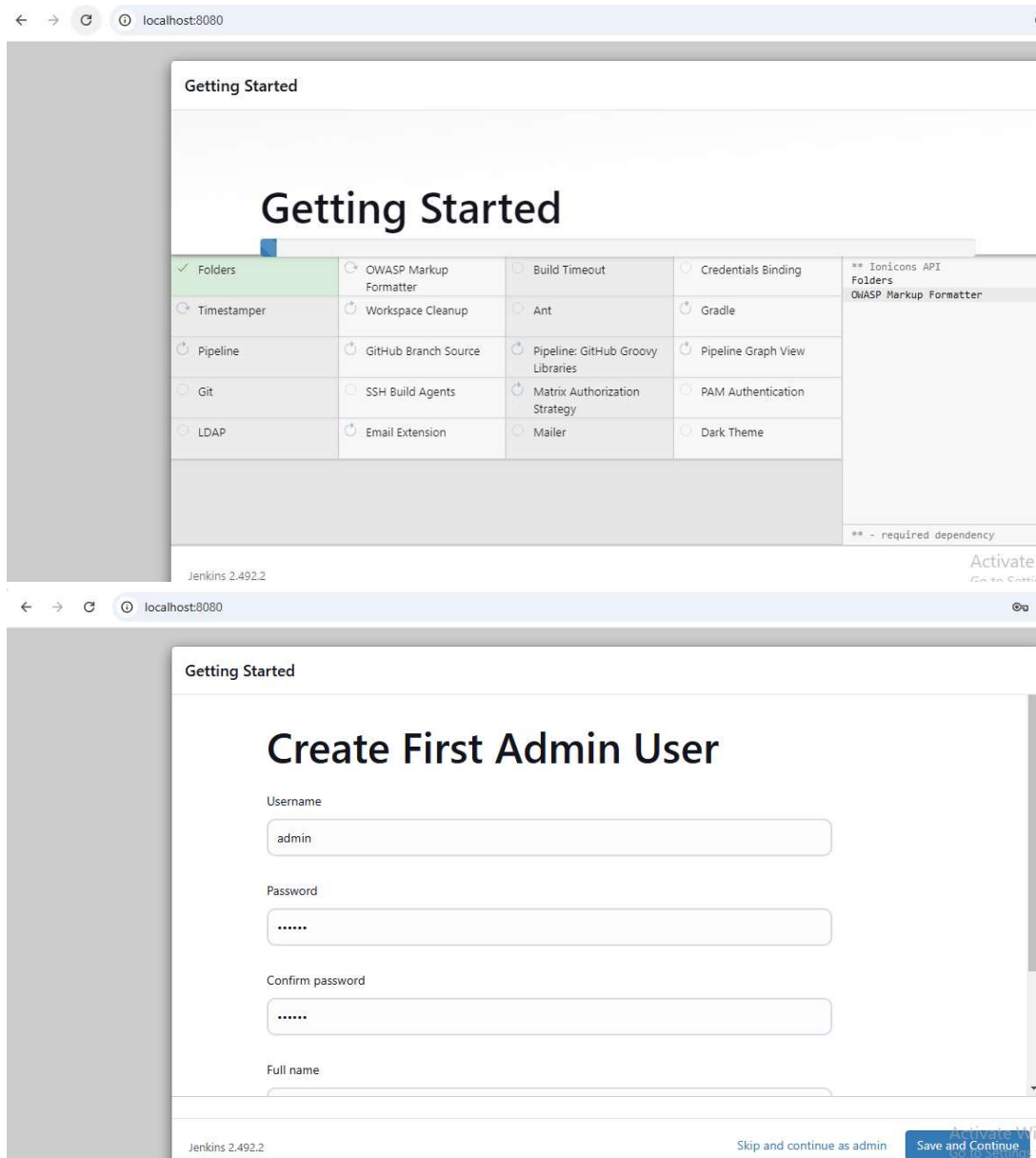
Plugins extend Jenkins with additional features to support many different needs.

Install suggested plugins

Install plugins the Jenkins community finds most useful.

Select plugins to install

Select and install plugins most suitable for your needs.



17. Jenkins is now installed and running inside WSL Ubuntu. Access the Jenkins dashboard from WSL instance

Getting Started

Instance Configuration

Jenkins URL:

<http://localhost:8080/>

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the BUILD_URL environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

[Activate Windows](#)

18. Click on save and finish

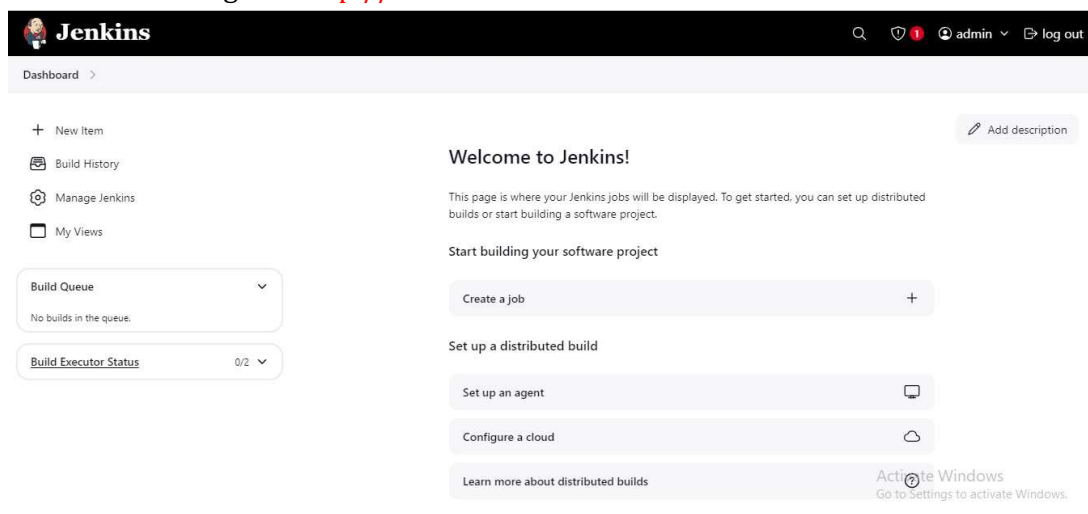
Getting Started

Jenkins is ready!

Your Jenkins setup is complete.

[Start using Jenkins](#)

19. Then click on start using Jenkins and the following dashboard will be displayed or Access Jenkins in Browser and go to: <http://localhost:8080>



The screenshot shows the Jenkins web interface. At the top is a black header with the Jenkins logo, a search icon, a shield icon with a red exclamation mark, a user dropdown menu showing 'admin', and a 'log out' link. Below the header is a 'Dashboard' section with a left sidebar containing links: '+ New Item', 'Build History', 'Manage Jenkins', and 'My Views'. The main content area has a 'Welcome to Jenkins!' message, followed by a paragraph explaining the dashboard's purpose. Below this is a section titled 'Start building your software project' with a 'Create a job' button. Another section titled 'Set up a distributed build' contains buttons for 'Set up an agent', 'Configure a cloud', and a link to 'Learn more about distributed builds'. On the left, there are two status boxes: 'Build Queue' showing 'No builds in the queue' and 'Build Executor Status' showing '0/2'. An 'Add description' link is in the top right of the main content area. At the bottom right, there is a Windows watermark: 'Activate Windows Go to Settings to activate Windows.'

Congratulations!! You have successfully installed jenkins on wsl Ubuntu.

3. Installing Jenkins on linux virtual machine

Prerequisites

- A Linux system with ubuntu installed on wsl
- A user account with sudo or root privileges
- Access to a terminal window/command line
- Java version 8

Tasks

Step 1: Install Java

Login in as a root user

```
sudo su
```

Install Java 8 on your system. If you already have Java 8 installed and set as the default, skip ahead to Step 2.

Open a terminal window, and enter the following: `yum install`

```
java-11-openjdk-devel -y
```

 To check the java version

```
java -version
```

```
paul@localhost:/home/paul
File Edit View Search Terminal Help
[paul@localhost ~]$ su
Password:
[root@localhost paul]# yum install java-11-openjdk-devel -y
Last metadata expiration check: 1 day, 6:23:10 ago on Thu 21 Sep 2023 02:40:47 PM IST.
Package java-11-openjdk-devel-1:11.0.20.0.8-3.0.1.el8.x86_64 is already installed.
Dependencies resolved.
Nothing to do.
Complete!
[root@localhost paul]# java --version
openjdk 11.0.20 2023-07-18 LTS
OpenJDK Runtime Environment (Red_Hat-11.0.20.0.8-2.0.1) (build 11.0.20+8-LTS)
OpenJDK 64-Bit Server VM (Red_Hat-11.0.20.0.8-2.0.1) (build 11.0.20+8-LTS, mixed mode, sharing)
[root@localhost paul]#
```

Step 2: Add Jenkins Software Repository

Jenkins isn't included in the default CentOS software repositories. To add the Jenkins repository, open a terminal window, and enter the following:

```
sudo wget -O /etc/yum.repos.d/jenkins.repo \
https://pkg.jenkins.io/redhat-stable/jenkins.repo
```

Import the GPG key to ensure your software is legitimate:

```
rpm --import https://pkg.jenkins.io/redhat-stable/jenkins.io-2023.key
```

If the process is successful, the system returns a new command line without error.

Step 3: Install Jenkins on CentOS 8

To Install Jenkins on CentOS 8, enter the following in the terminal:

```
yum install Jenkins* -y
```

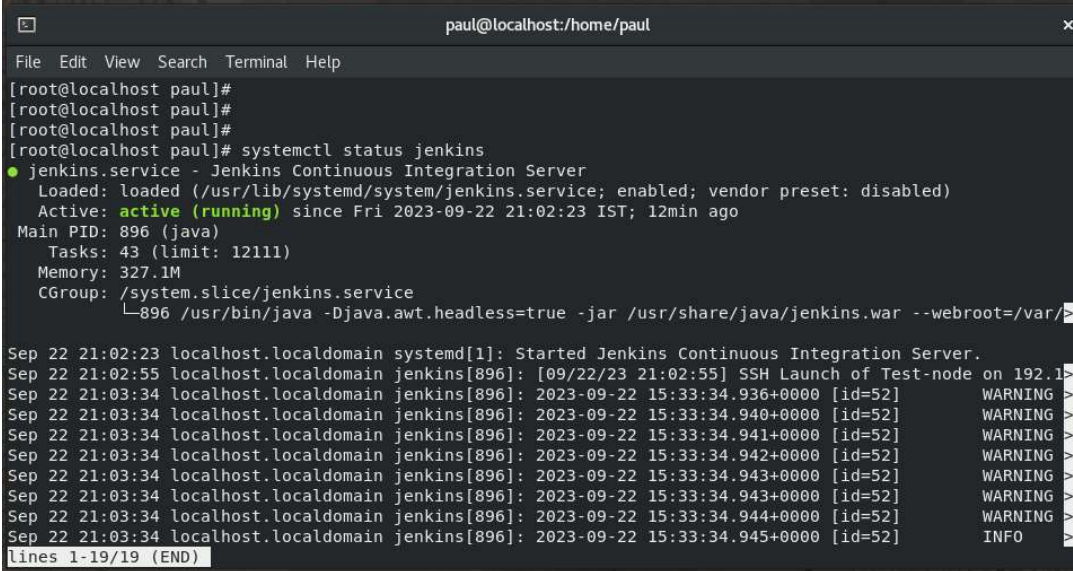
Start Jenkins Service

To start the Jenkins service and enable it at startup, enter the following:

```
sudo systemctl start jenkins sudo  
systemctl enable jenkins
```

To display the status of the Jenkins service, enter the following:

```
sudo systemctl status Jenkins
```

A terminal window titled 'paul@localhost:/home/paul' showing the command 'systemctl status jenkins' and its output. The output indicates that the Jenkins service is active and running. Below the status, a series of log messages are shown, including the start of the Jenkins Continuous Integration Server and several warnings from the jenkins[896] process.

```
paul@localhost:/home/paul  
File Edit View Search Terminal Help  
[root@localhost paul]#  
[root@localhost paul]#  
[root@localhost paul]#  
[root@localhost paul]# systemctl status jenkins  
● jenkins.service - Jenkins Continuous Integration Server  
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; vendor preset: disabled)  
   Active: active (running) since Fri 2023-09-22 21:02:23 IST; 12min ago  
 Main PID: 896 (java)  
    Tasks: 43 (limit: 12111)  
  Memory: 327.1M  
   CGroup: /system.slice/jenkins.service  
           └─896 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/>  
Sep 22 21:02:23 localhost.localdomain systemd[1]: Started Jenkins Continuous Integration Server.  
Sep 22 21:02:55 localhost.localdomain jenkins[896]: [09/22/23 21:02:55] SSH Launch of Test-node on 192.1>  
Sep 22 21:03:34 localhost.localdomain jenkins[896]: 2023-09-22 15:33:34.936+0000 [id=52] WARNING >  
Sep 22 21:03:34 localhost.localdomain jenkins[896]: 2023-09-22 15:33:34.940+0000 [id=52] WARNING >  
Sep 22 21:03:34 localhost.localdomain jenkins[896]: 2023-09-22 15:33:34.941+0000 [id=52] WARNING >  
Sep 22 21:03:34 localhost.localdomain jenkins[896]: 2023-09-22 15:33:34.942+0000 [id=52] WARNING >  
Sep 22 21:03:34 localhost.localdomain jenkins[896]: 2023-09-22 15:33:34.943+0000 [id=52] WARNING >  
Sep 22 21:03:34 localhost.localdomain jenkins[896]: 2023-09-22 15:33:34.943+0000 [id=52] WARNING >  
Sep 22 21:03:34 localhost.localdomain jenkins[896]: 2023-09-22 15:33:34.944+0000 [id=52] WARNING >  
Sep 22 21:03:34 localhost.localdomain jenkins[896]: 2023-09-22 15:33:34.945+0000 [id=52] INFO >  
lines 1-19/19 (END)
```

Step 4: Set Firewall to Allow Jenkins

To add port 8080 in firewall

```
firewall-cmd --add-port=8080/tcp --permanent
```

To reload the firewall

```
firewall-cmd --reload
```

To list the default zone to check the port

```
firewall-cmd --list-all
```

Check the ip address of your vm

```
ip addr show
```

Step 5: Run and Set up Jenkins on CentOS 8

Next, you'll test your system to make sure Jenkins is working correctly. Open a web browser, and enter the following URL:

Ip address:8080

The browser should display an **Unlock Jenkins** page. It will ask you to enter a temporary password.



This password was created automatically during setup. To find it, switch to a terminal window and enter the following:

```
sudo cat  
/var/lib/jenkins/secrets/initialAdminPassword
```

The location is displayed on the **Getting Started / Unlock Jenkins** page.

The system will display an alphanumeric code. Copy this code and paste it into the password field. Then click **Continue**.

Jenkins will prompt you to install plugins. It is recommended that you **Install suggested plugins**. You can always change or customize plugins later.

Once plugins are configured, you'll be prompted to **Create First Admin User**. Enter the values you want for your Jenkins Administrator, then click **Save and Continue**.

The last page will display the **Instance Configuration**. This shows the hostname Jenkins is configured for. This is usually the same hostname you typed into your browser to access the **Getting Started** page. If you do not create an admin user, there will be a warning.

Click **Save and Finish** to start using Jenkins.

Getting Started

Instance Configuration

Jenkins URL:

The Jenkins URL is used to provide the root URL for absolute links to various Jenkins resources. That means this value is required for proper operation of many Jenkins features including email notifications, PR status updates, and the `AUTO_URL` environment variable provided to build steps.

The proposed default value shown is **not saved yet** and is generated from the current request, if possible. The best practice is to set this value to the URL that users are expected to use. This will avoid confusion when sharing or viewing links.

Jenkins 2.346.1

Not now **Save and Finish**

This step takes you to the Jenkins dashboard where you can create new jobs.

- + New Item
-  Build History
-  Manage Jenkins
-  My Views

Build Queue ▾

No builds in the queue.

Build Executor Status 0/2 ▾

Welcome to Jenkins!

 Add description

This page is where your Jenkins jobs will be displayed. To get started, you can set up distributed builds or start building a software project.

Start building your software project

Create a job +

Set up a distributed build

Set up an agent 

Configure a cloud 

Learn more about distributed builds

Activate Windows
Go to Settings to activate Windows.